7. Source Code for the LeaderRank algorithm

We attached here the source code for running LeaderRank algorithm:

```
% This is a Matlab M-file for LeaderRank.
E=load('Network.dat'); % Read the network data with different pairs
                       % of 'fan leader' in consecutive rows, and the
                       % labels of nodes should start from 1
l=length(E);           % l is the number of links
N=max(max(E));         % N is the number of nodes.

% Add ground node and create adjacency matrix P
    EG1=zeros(N,2);
    EG2=zeros(N,2);
    for i=1:N
        EG1(i,1)=N+1;
        EG1(i,2)=i;
    end
    EG2(:,1)=EG1(:,2);
    EG2(:,2)=EG1(:,1);
    E=[E;EG1;EG2];
    P=sparse(E(:,1),E(:,2),1);
    D_in=sum(P);       % in degree
    D_out=sum(P');     % out degree

% Transition matrix PP
    EE=zeros(N+1,2);
    for j=1:N+1
        EE(j,1)=j;
        EE(j,2)=1/D_out(j);
    end
    D=sparse(EE(:,1),EE(:,1),EE(:,2));
    PP=D*P;

% Diffusion to stable state.
    God=zeros(N+1,1);
    God(1:N,1)=1;                  % Assign initial resource
    error=10000;                   % error is the average error of nodes' scores.
    error_threshold=0.00002;       % It is a tunable parameter controlling the
                                   % error tolerance.
    step=1;
    while error>error_threshold
        step
        M=God;
        God=PP'*God;
        error=sum(abs(God-M)./M)/(N+1);
        step=step+1;
    end
    b=God(N+1)/N;
    God=God+b;
    God(N+1)=0;

% Write the ranking results to "Results.dat": node's ID & Score
    R=zeros(N,2);
    R(:,1)=[1:N]';
```

```
R(:,2)=God(1:N);
[ b, pos ] = sort( -R( :, 2 ));
R = R(pos,: );
fid = fopen('Results.dat','w');
for i=1:N
    fprintf(fid,'%d %f \n',R(i,1),R(i,2));
end
fclose(fid);
```